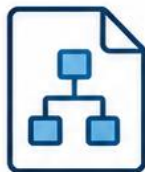


HIGH LEVEL DESIGN (HLD) & LOW LEVEL DESIGN (LLD)

End-to-End E-Commerce Analytics Platform



Project:
**End-to-End E-Commerce
Analytics Platform**

Prepared By:

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Dataset:

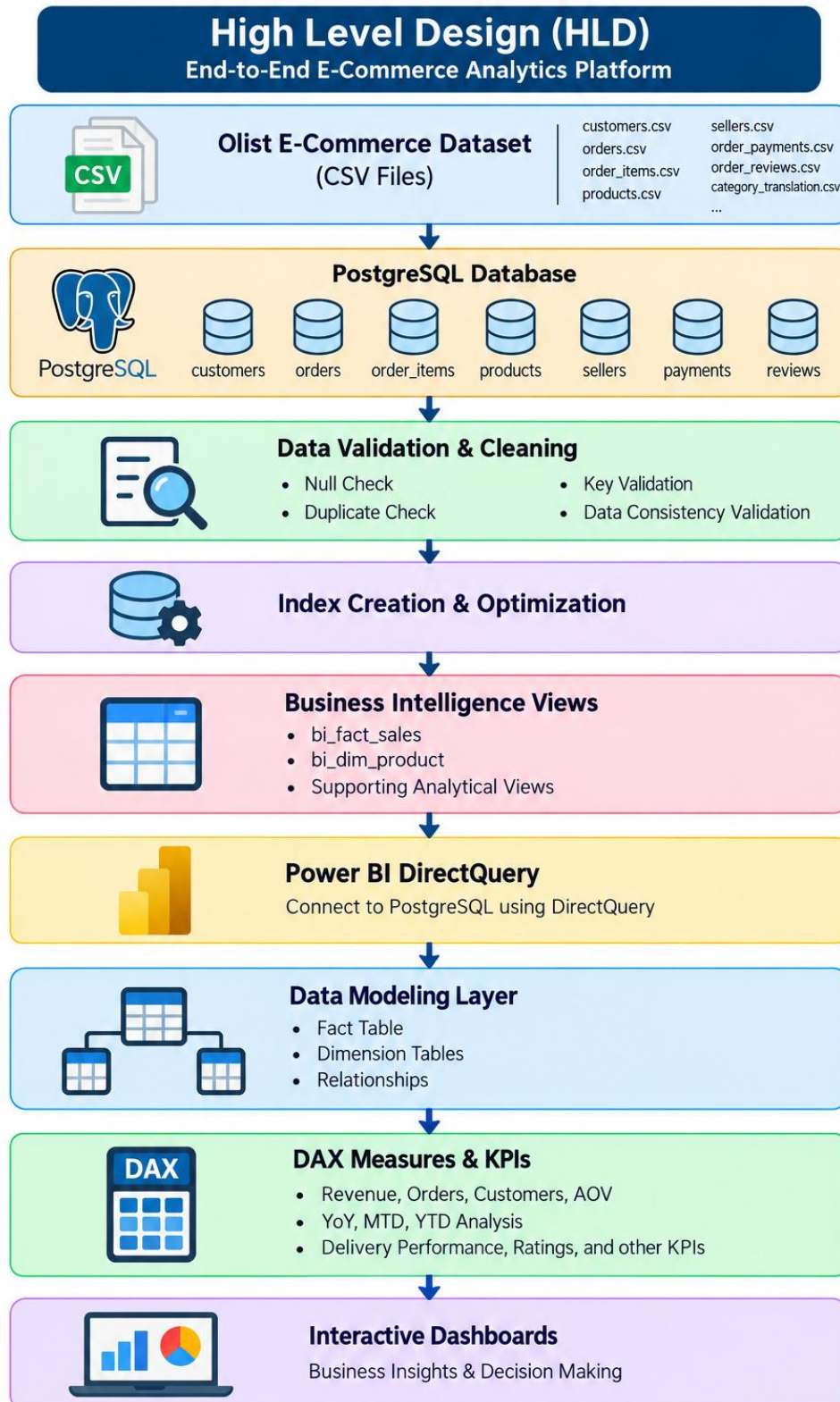
Olist E-Commerce Dataset

(CSV Files)

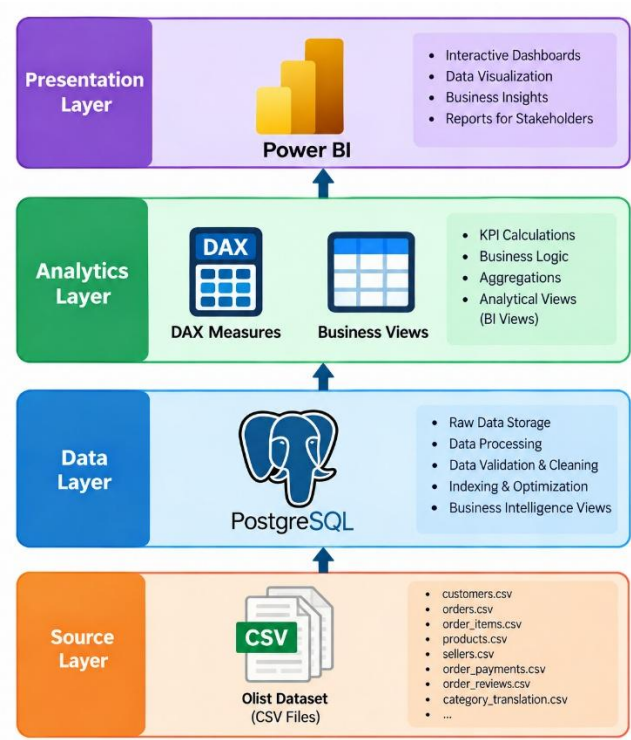
HIGH LEVEL DESIGN (HLD)

The End-to-End E-Commerce Analytics Platform is a Business Intelligence solution designed to transform raw e-commerce transaction data into meaningful business insights. The project uses the Olist E-Commerce Dataset containing customer, order, product, seller, payment, and review information.

The High-Level Design describes how data flows from source files to dashboards.



Technology Architecture



Major Components

Component	Purpose
Dataset Layer	Stores source CSV files
Database Layer	Stores raw transactional data
Validation Layer	Ensures data quality
Optimization Layer	Improves query performance
BI View Layer	Creates reporting datasets
Analytics Layer	KPI calculations
Visualization Layer	Dashboard generation

HLD Conclusion

The High-Level Design (HLD) of the End-to-End E-Commerce Analytics Platform provides a comprehensive overview of how data flows from the source dataset to the final business intelligence dashboards. The architecture is designed to ensure efficient data storage, validation, transformation, and reporting by leveraging PostgreSQL for data management and Power BI for visualization. Through the implementation of optimized database structures, analytical views, DirectQuery connectivity, and KPI-driven dashboards, the system enables stakeholders to gain actionable insights into sales performance, customer behavior, product trends, logistics efficiency, seller contribution, and customer satisfaction. The modular and scalable architecture also provides a strong foundation for future enhancements such as cloud deployment, real-time analytics, predictive modeling, and advanced business intelligence capabilities.

LOW LEVEL DESIGN (LLD)

Olist E-Commerce Analytics Platform - Database Layer, BI Views, Data Model & Dashboard Architecture

1. Database Layer

Raw transactional tables imported from the Olist dataset.

Customers Table

Stores customer information.

`customer\_id` `customer\_unique\_id` `customer\_city` `customer\_state`

[\[Customer Analysis\]](#) [\[Geographic Analysis\]](#)

Orders Table

Stores order lifecycle information.

`order\_id` `customer\_id` `order\_status` `order\_purchase\_timestamp`

`order\_delivered\_customer\_date` `order\_estimated\_delivery\_date`

[\[Revenue Tracking\]](#) [\[Delivery Analysis\]](#)

Order Items Table

Stores individual products within orders.

`order\_id` `product\_id` `seller\_id` `price` `freight\_value`

[\[Product Analysis\]](#) [\[Seller Analysis\]](#)

Products Table

Stores product metadata.

`product\_id` `product\_category\_name` `product\_weight\_g` `product\_length\_cm`

`product\_height\_cm` `product\_width\_cm`

[\[Category Analysis\]](#)

Sellers Table

Stores seller information.

`seller\_id` `seller\_city` `seller\_state`

[\[Seller Performance Analysis\]](#)

Payments Table

Stores payment information.

`payment\_type` `payment\_installments` `payment\_value`

[\[Payment Analysis\]](#)

Reviews Table

Stores customer feedback.

`review\_score` `review\_comment\_message`

[\[Customer Satisfaction Analysis\]](#)

2. Business Intelligence Views

bi_fact_sales

Central reporting dataset — the primary fact table used in Power BI.

Combines:

`Orders` `Customers` `Products` `Order Items` `Payments` `Reviews` `Sellers`

Business Metrics:

`Revenue` `Orders` `Customers` `Delivery Days` `Ratings` `Payments`

bi_dim_product

Product dimension.

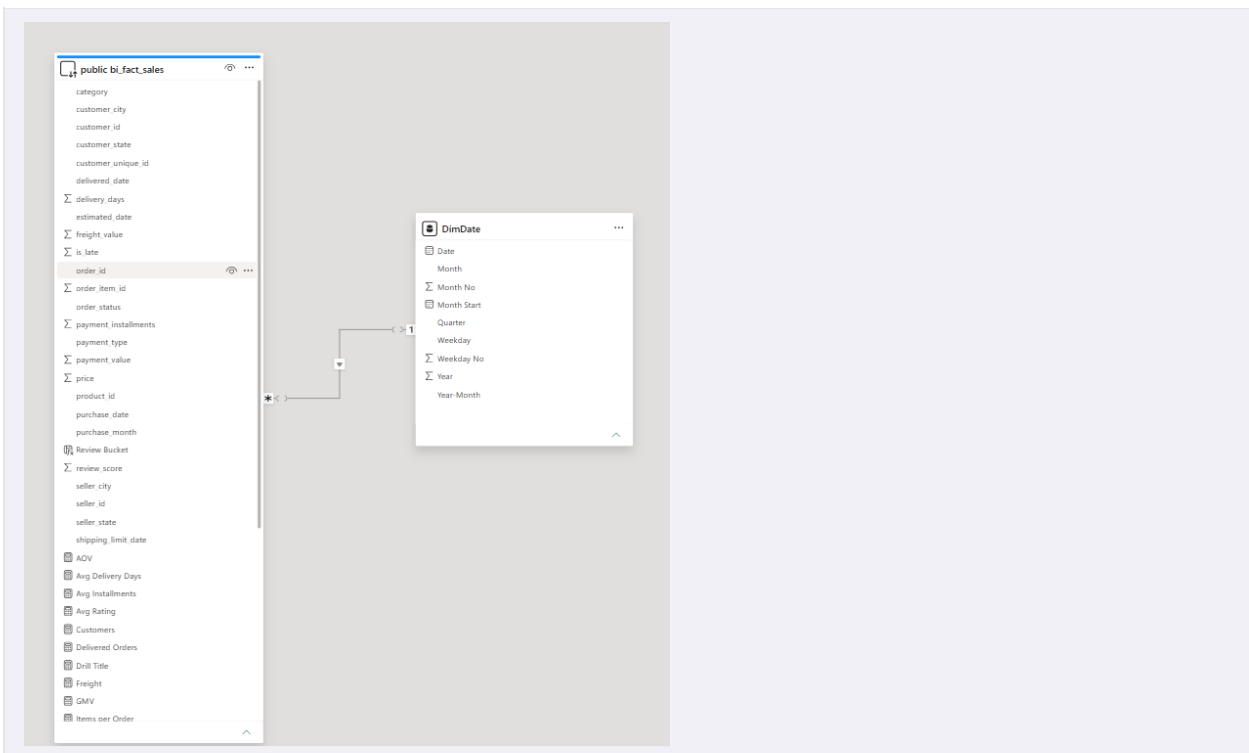
Contains:

`Product Category` `Product Attributes` `Category Translation`

Used for:

`Category Analysis` `Product Analysis`

3. Power BI Data Model



4. DAX Measures

```
Revenue =  
SUM(bi_fact_sales[revenue])
```

```
Orders =  
DISTINCTCOUNT(bi_fact_sales[order_id])
```

```
Customers =  
DISTINCTCOUNT(bi_fact_sales[customer_unique_id])
```

```
AOV =  
DIVIDE([Revenue],[Orders])
```

```
Avg Rating =  
AVERAGE(bi_fact_sales[review_score])
```

```
On Time % =  
DIVIDE([On Time Orders],[Orders])
```

5. Dashboard Architecture

Executive Overview Page

KPIs:

- Revenue
- Orders
- Customers
- AOV
- Avg Rating
- On-Time %

Provide management-level overview.

Sales Trends Page

Visuals:

- Revenue Trend
- Monthly Sales
- Revenue by Category

Track sales performance.

Product & Category Analysis Page

Visuals:

- Top Products
- Category Performance

Identify product trends.

Customer Analysis Page

Visuals:

- Customer Distribution
- Revenue by State

Understand customer behavior.

Delivery & Logistics Analysis Page

Visuals:

- Delivery Time
- Delayed Orders

Monitor logistics efficiency.

Reviews Analysis Page

Visuals:

- Rating Distribution
- Review Trends

Measure customer satisfaction.

Payments Analysis Page

Visuals:

- Payment Type Distribution
- Installment Analysis

Understand payment behavior.

Seller Analysis Page

Visuals:

- Seller Revenue
- Seller Ranking

Evaluate seller performance.

LLD Conclusion

The Low-Level Design (LLD) provides a detailed view of the system's internal implementation, including database tables, relationships, BI views, data models, and KPI calculations. It defines how data is processed, transformed, and presented within the analytics platform. The design ensures consistency, maintainability, and efficient reporting across all business modules. This document serves as a technical reference for future development, enhancements, and system maintenance.